

Project 9 — Pathway & Functional Enrichment Analysis

Interpreting transcriptomic signals associated with chemotherapy response in breast cancer

Research question: What biological processes and signaling pathways are associated with transcriptomic differences between chemotherapy-sensitive and chemotherapy-resistant breast cancer patients?

1. Background

This project builds on Project 8 and moves from a differential-expression gene list toward biological interpretation. Functional enrichment asks whether genes associated with a phenotype are concentrated in particular biological processes or pathways.

2. Dataset

Resource	GSE162187 / PRJNA680808
Organism	Homo sapiens
Experiment	RNA-seq
Patient profiles	22
Sensitive	9
Resistant	13
DEGs reported	1,985

The original study used Kallisto for transcript quantification, DESeq2 for differential expression, and Benjamini–Hochberg correction. GO and KEGG enrichment were performed on DEGs with $P < 0.05$; the study also used GSEA and STRING-based protein association analysis.

3. Gene Ontology findings

The published GO analysis identified biological themes including cell and tissue development, extracellular-matrix assembly, positive regulation of epithelial-to-mesenchymal transition (EMT), regulation of cellular response to TGF- β receptor signaling, homophilic cell adhesion, epithelium development, regulation of angiogenesis and cell migration.

These findings suggest that chemotherapy response is associated with coordinated changes in tissue architecture, epithelial phenotype, adhesion and migration rather than a single isolated gene.

4. KEGG pathway findings

The published KEGG analysis highlighted pathways related to transcriptional regulation, cell proliferation and signal transduction. Among the first 15 pathways were PI3K-Akt, MAPK, calcium, TGF- β , Ras, cAMP and JAK-STAT signaling.

PI3K-Akt and MAPK are particularly relevant because they participate in cell-survival and proliferation signaling. The paper also discusses links between MAPK activity, invasion, EMT and drug resistance. TGF- β is notable because it can promote EMT and invasive behavior in appropriate cancer contexts.

5. STRING functional clusters

The study filtered genes using adjusted $P < 0.05$ and examined protein associations with STRING using a high-confidence threshold of 0.7. Reported clusters included extracellular-matrix proteins, epithelial/keratin-associated proteins, plasma-membrane proteins, HLA proteins linked with chemical-stimulus response, and UGT proteins involved in drug metabolism.

6. Biological interpretation

Taken together, the published results point toward extracellular-matrix organization, epithelial biology, cell adhesion and migration, TGF- β -related signaling and several major cancer-associated signaling pathways as important themes in chemotherapy response. These pathways are useful hypotheses for follow-up research, but enrichment does not establish causality.

7. Reproducibility note

This portfolio project deliberately separates published findings from independently generated analysis. The complete count-level workflow and full DEG table were not independently regenerated here. Therefore, the numerical enrichment findings are attributed to the original peer-reviewed study rather than presented as newly generated statistics. The figures and explanatory organization in this portfolio are original.

8. Limitations

- The original cohort contains only 22 patients, limiting statistical power.
- Published enrichment results were not fully recomputed from raw/count-level data here.
- Enrichment indicates association rather than causation.
- A stronger follow-up would obtain the complete DEG table, reproduce GO/KEGG/GSEA computationally, document the background gene universe, and validate findings in an independent cohort.

9. Conclusion

Project 9 moves the portfolio from “which genes change?” to “what biology changes?” The study's enrichment results suggest coordinated involvement of extracellular-matrix biology, epithelial and adhesion processes, migration, TGF- β -related signaling and several major cancer-associated pathways. The next scientific step would be to reproduce these results computationally and test their stability in independent datasets.

10. Sources

Barrón-Gallardo CA, Garcia-Chagollán M, Morán-Mendoza AJ, et al. (2022). Transcriptomic Analysis of Breast Cancer Patients Sensitive and Resistant to Chemotherapy: Looking for Overall Survival and Drug Resistance Biomarkers. *Technology in Cancer Research & Treatment*, 21. DOI: 10.1177/15330338211068965.

NCBI GEO: GSE162187. NCBI BioProject: PRJNA680808.